BOOK CONTENT

HISTOLOGY - PROF. DALIA ELMARAKBY

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PRACTICE TIME

1. The electron microscopic picture of cell membrane

(A) Two-unit membranes separated by space.
(B) 3-10 flat curved membranes.
(C) Two dark lines separated by clear one.
(D) Two subunits small and large.

2. Protein components of the cell membrane is

(A) Peripheral present inside the lipid bilayer.
(B) Peripheral small continuous molecules outside the lipid bilayer.
(C) Integral present outside the lipid bilayer.
(D) Integral may cross the lipid bilayer.

3. Glycocalyx is formed of

(A) Molecules of lipoproteins and glycoproteins.
(B) Molecules of glycolipids and glycoproteins.
(C) Molecules of proteoglycans and glycoproteins.
(D) Molecules of phospholipids and cholesterol.

4. What is the limiting membrane of a cell?

(A) Plasmalemma. (B) Glycocalyx. (C) Protoplasm. (D) Cristae.

5. The following stains can be used to stain the cell membrane

(A) Silver stain. (B) Trichrome stains. (C) Orcein stain. (D) Indian ink.

6. As regard the ribosomes one of the following is correct:

(A) The rRNA is formed in the pars granulosa & combined with the protein formed in the cytoplasm.
(B) The rRNA is formed in the nucleolus & combined with the protein formed in the nucleus.
(C) The rRNA is formed in the nucleolus & combined with the protein formed in the nucleolus.
(D) The rRNA is formed in the pars amorpha & combined with the protein formed in the cytoplasm.

7. The muscle cells could contain all the following filaments except:

(A) Actin. (B) Vimentin. (C) Keratin. (D) Desmin.

8. As regard the microtubules one of the following is incorrect:

(A) They are formed of tubulin protein by polymerizations.
(B) They arise from the microtubular organizing center.
(C) They have unfixed length.
(D) They share in the formation of centrioles, cilia, microvilli and flagella.

9. As regard the microvilli one of the following is correct:

(A) They are formed of core of microtubules covered with cell membrane.
(B) They are formed of core microtubules & not covered with cell membrane.
(C) They are formed of core of microfilaments covered with cell membrane.
(D) They are formed of core of actin filaments & not covered with cell membrane.

1. c 2. d 3. b 4. a 5. a 6. a 7. c 8. d 9. c

10. In dwarfism the following could be the cause except:

(A) Defect in the formation of the glycoprotein growth hormone in the rER.
(B) Defect in the formation of the glycoprotein growth hormone in the sER.
(C) Decreased amount of the secreted growth hormone.
(D) Defect in the growth hormone' receptors on the target cells.

11. In glycogen storage diseases the cause could be:

(A) Defect in the rER.
(B) Defect in the sER.
(C) Defect in the lysosomes.
(D) Defect in the peroxisomes.

12. All of the following organelles could share in the formation of a protein hormone except:

(A) Rough endoplasmic reticulum.
(B) Golgi apparatus.
(C) Mitochondria.
(D) Smooth endoplasmic reticulum.

13. One of the following is correct as regard the inner nuclear membrane; it is linked to:

(A) Ribosomes.
(B) Myosin filaments.
(C) Lamins filaments.
(D) Euchromatin filaments.

14. Which of the following is NOT a membranous organelle?

(A) Lysosomes. (B) Peroxisomes. (C) Mitochondria. (D) Ribosomes.

15. Which organelle contains digestive enzymes?

(A) Rough endoplasmic reticulum.
(B) Smooth endoplasmic reticulum.
(C) Lysosome.
(D) Mitochondria.

16. What structure is at the base of cilia?

(A) Ribosomes. (B) Peroxisome. (C) Microfilaments. (D) Centrioles.

17. Which of the following is NOT considered an inclusion?

(A) Pigment. (B) Glycogen. (C) Lipid. (D) Peroxisome.

18. How thick is the plasma membrane?

(A) 7.5–10 angstroms.
(B) 7.5–10 nanometers.
(C) 7.5–10 micrometers.
(D) 7.5–10 millimeters.

19. What is the term for the general process that cells use to extrude material from the cell?

(A) Endocytosis. (B) Exocytosis. (C) Pinocytosis. (D) Phagocytosis.

10. b 11. b 12. d 13. c 14. d 15. c 16. d 17. d 18. b 19. b

20. Which organelle stores, concentrates and packages proteins within a cell?

Ⓐ Rough endoplasmic reticulum.
Ⓑ Smooth endoplasmic reticulum.
Ⓒ Lysosome.
Ⓓ Golgi apparatus.

21. Which of the following is NOT a membranous organelle?

Ⓐ Lysosomes. Ⓑ Filaments. Ⓒ Peroxisomes. Ⓓ Mitochondria.

22. What are the folds on the inner mitochondrial membrane called?

Ⓐ Matrix. Ⓑ Saccules. Ⓒ Cisternae. Ⓓ Cristae.

23. Which organelle is not involved in lipid metabolism?

Ⓐ Rough endoplasmic reticulum.
Ⓑ Smooth endoplasmic reticulum.
Ⓒ Peroxisomes.
Ⓓ Golgi apparatus.

24. Which organelle contains detoxifying enzymes?

Ⓐ Ribosomes. Ⓑ Peroxisome. Ⓒ Microfilaments. Ⓓ Centrioles.

25. Which of the following is NOT a membranous organelle?

Ⓐ Microtubules. Ⓑ Lysosomes. Ⓒ Peroxisomes. Ⓓ Mitochondria.

26. What is the term for the general process that cells us to bring things into the cell?

Ⓐ Endocytosis.
Ⓑ Exocytosis.
Ⓒ Receptor mediated endocytosis.
Ⓓ Phagocytosis.

27. Which organelle produces protein for export?

Ⓐ Rough endoplasmic reticulum.
Ⓑ Smooth endoplasmic reticulum.
Ⓒ Lysosome.
Ⓓ Ribosomes.

28. Where are ribosomal subunits constructed?

Ⓐ Cytoplasm.
Ⓑ Pars amorpha.
Ⓒ Rough endoplasmic reticulum.
Ⓓ Nucleolus.

29. What part of the cell is responsible for breaking down and digesting things?

Ⓐ Ribosomes.
Ⓑ Lysosomes.
Ⓒ Endoplasmic reticulum.
Ⓓ Vacuole.

30. What part of the cell serves as the intracellular pass way?

Ⓐ Endoplasmic reticulum.
Ⓑ Golgi apparatus.
Ⓒ Cell membrane.
Ⓓ Mitochondria.

20. d 21. b 22. d 23. a 24. b 25. a 26. a 27. a 28. d 29. b 30. a

31. The jellylike interior of the cell is called the:

Ⓐ Vacuole. Ⓑ Cytosol. Ⓒ Cytoskeleton. Ⓓ Nucleus.

32. What part of the cell makes proteins?

Ⓐ Ribosomes. Ⓑ Mitochondria. Ⓒ Lysosomes. Ⓓ Vacuole.

33. The inner mitochondrial membrane are connected to globular structures that contain:

Ⓐ Citric acid cycle enzymes.
Ⓑ ATP synthetase.
Ⓒ Reductase enzymes.
Ⓓ Oxidoreductase enzymes.

34. Which statement is correct:

Ⓐ Peroxisomes contain hydrolytic enzymes.
Ⓑ Mitochondria contain oxidoreductase enzymes.
Ⓒ Lysosomes contain oxidative enzymes.
Ⓓ Microbodies contain catalase enzymes.

35. Both oxidases and catalase enzymes are present in:

Ⓐ Mitochondria. Ⓑ Microtubules. Ⓒ Microbodies. Ⓓ Microfilaments.

36. Which of the following statements is correct:

Ⓐ Cell membrane thickness is 7.5–10 µm.
Ⓑ Microtubules' diameter is 24 mm.
Ⓒ Microfilaments' diameter is 5–7 nm.
Ⓓ Microfilaments is 5–10 nm in length.

37. Regarding primary lysosome, the following is true:

Ⓐ Increase in protein forming cell.
Ⓑ Can appear by histochemical test.
Ⓒ Is heterogeneous by EM.
Ⓓ Contain respiratory enzymes.

38. The secretory vesicles arise from the:

Ⓐ Tubules of rER.
Ⓑ Cis face of the Golgi apparatus.
Ⓒ Tubules of sER.
Ⓓ Trans face of the Golgi apparatus.

39. Shaft of cilia contains:

Ⓐ Peripheral 9 triplets+2 central singlets.
Ⓑ Peripheral 9 doublets+2 central singlets.
Ⓒ Peripheral 9 tubules+2 central singlets.
Ⓓ Peripheral 9 triplets.

40. Basal body of cilia is similar to:

Ⓐ Centriole. Ⓑ Centrosome. Ⓒ Centromere. Ⓓ Ribosome.

41. Euchromatic nucleus has the following except:

Ⓐ Active.
Ⓑ Pale.
Ⓒ Condensed.
Ⓓ Prominent nucleolus.

31. b 32. a 33. b 34. d 35. c 36. c 37. b 38. d 39. b 40. a 41. c

42. Iron hematoxylin used to stain:

(A) Mitochondria and Golgi.
(B) Golgi and centriole.
(C) Mitochondria and centriole.
(D) Mitochondria and rough endoplasmic reticulum.

43. PAS stain is used to demonstrate all of the following except:

(A) Cell membrane.
(B) Glycogen.
(C) Fat.
(D) Basement membrane.

44. Silver stain is used to demonstrate:

(A) Golgi.
(B) Fat.
(C) Cell membrane.
(D) a and c.

45. Nucleolar organizer is:

(A) Pars fibrosa.
(B) Early formed rRNA.
(C) DNA.
(D) Granular form of RNA.

46. Which of the following organelles share in lipid metabolism:

(A) Smooth endoplasmic reticulum.
(B) Ribosome.
(C) Peroxisome.
(D) a and c.

47. One of the following is NOT an intermediate filament:

(A) Lamins.
(B) Actin.
(C) Vimentin.
(D) Keratin.

48. Secondary lysosomes include the following except:

(A) Multivesicular bodies.
(B) Heterolysosomes.
(C) Residual bodies.
(D) Autolysosome.

49. What structure contributes to the cells cytoskeleton?

(A) Ribosomes.
(B) Peroxisome.
(C) Microfilaments.
(D) Centrioles.

50. Heterolysosome is formed of:

(A) Primary lysosome + pinocytic vesicle.
(B) Primary lysosome + phagosome.
(C) Primary lysosome + autophagic vesicle.
(D) Primary lysosome + Residual body.

51. The nucleus of protein forming cell shows all the following except:

(A) More euchromatin by EM.
(B) More heterochromatin by EM.
(C) Clear nucleolus.
(D) Pale by LM.

52. Regarding the cell membrane, the following is true:

(A) Clearly appears by LM.
(B) Its main function is cell respiration.
(C) Can be stained by H & E.
(D) Appear by EM as triple layers.

42. c 43. c 44. d 45. c 46. d 47. b 48. c 49. c 50. b 51. b 52. d

53. All of the following are membranous organelles except:

Ⓐ Cell membrane.
Ⓑ Rough endoplasmic reticulum.
Ⓒ Centriole.
Ⓓ Lysosome.

54. The most predominant organelle in lipid producing cell is:

Ⓐ Golgi complex.
Ⓑ sER.
Ⓒ Lysosomes.
Ⓓ rER.

55. What process is nicknamed "cell eating"?

Ⓐ Pinocytosis.
Ⓑ Active transport.
Ⓒ Endocytosis.
Ⓓ Phagocytosis.

56. An organelle in the cytoplasm that has DNA is called:

Ⓐ rER.
Ⓑ Mitochondria.
Ⓒ Lysosomes.
Ⓓ Golgi apparatus.

57. The transfer vesicles arise from the:

Ⓐ Tubules of rER.
Ⓑ Ribosomes.
Ⓒ Golgi apparatus.
Ⓓ Tubules of sER.

58. The cytoplasmic surface of the pit is coated by

Ⓐ Vimentin.
Ⓑ Keratin.
Ⓒ Clathrin.
Ⓓ Desmin.

59. Endoplasmic reticulum is formed of

Ⓐ Flat curved membranous saccules.
Ⓑ Continuous interconnected cisternae.
Ⓒ Vesicles with two membranes.
Ⓓ Granules with two subunits.

60. Basophilia of the cytoplasm is due to

Ⓐ Ribosomes and rER.
Ⓑ Smooth endoplasmic reticulum.
Ⓒ Golgi apparatus.
Ⓓ Lysosomes.

61. Initial glycosylation is a function of

Ⓐ sER.
Ⓑ Lysosomes.
Ⓒ Golgi apparatus.
Ⓓ rER.

62. Ribosomes attached to the rER by

Ⓐ Desmin.
Ⓑ Vimentin.
Ⓒ Ribophorin.
Ⓓ Lamin.

63. In H&E-stained sections of protein forming cell, Golgi appears as:

Ⓐ Deeply basophilic area near the nucleus.
Ⓑ Deeply acidophilic area near the nucleus.
Ⓒ Brown fibrils and granules near the nucleus.
Ⓓ Unstained area near the nucleus.

53. c 54. b 55. d 56. b 57. a 58. c 59. b 60. a 61. d 62. c 63. d

64. Cis face of Golgi

(A) Vesicles bud off from it.
(B) Receives vesicles from rER.
(C) Connects the saccules together.
(D) Fuse with cell membrane.

65. Renewal of the cell membrane is a function of

(A) sER.
(B) rER.
(C) Golgi apparatus.
(D) Lysosomes.

66. The outer mitochondrial membrane

(A) Forms cristae.
(B) Shows elementary particles.
(C) Smooth with porins.
(D) Shows attached ribosomes.

67. The inner mitochondrial membrane are connected to globular structures that contain

(A) Citric acid cycle enzymes.
(B) Calcium granules.
(C) Proteins with ATP synthetase activity.
(D) Circular DNA molecules.

68. Golgi is apical in position in

(A) Liver cells.
(B) Steroid secreting cells.
(C) Secretory cells.
(D) Nerve cells.

69. The euchromatin

(A) Contains inactive genes.
(B) Appears dark in EM.
(C) Predominates in active cells.
(D) Include peripheral and islands chromatin.

70. Where are the enzymes of Kreb's cycle located?

(A) Outer mitochondrial membrane.
(B) Inner mitochondrial membrane.
(C) Matrix granules.
(D) Mitochondrial matrix.

71. The resolution of TEM is about

(A) 0.2 nm.
(B) 0.4 µm.
(C) 0.2 mm.
(D) 0.4 mm.

72. Clumps of nucleoprotein concentrated near the periphery of the nucleus are called

(A) Nuclear pore complex.
(B) Nucleolus.
(C) Heterochromatin.
(D) Euchromatin.

73. Which structures consist of actin?

(A) Flagella.
(B) Cilia.
(C) Microfilaments.
(D) Microtubules.

74. A lysosome measures 0.4 µm in diameter. What is the diameter in nm?

(A) 4 nm.
(B) 40 nm.
(C) 400 nm.
(D) 4000 nm.

64. b 65. c 66. c 67. c 68. c 69. c 70. d 71. a 72. c 73. c 74. c

75. Enzymes associated with drug detoxification are associated with:

(A) sER.
(B) rER.
(C) Golgi body.
(D) Inner mitochondrial membrane.

76. Nucleolonema is:

(A) Early rRNA & mature rRNA.
(B) DNA & Early rRNA.
(C) DNA & mature rRNA.
(D) All the above.

77. All of the following are basophilic except:

(A) Nuclear membrane
(B) Chromatin.
(C) rER.
(D) Mitochondria.

78. Microbodies are:

(A) Peroxisomes.
(B) Lysosomes.
(C) Ribosomes.
(D) Centrosomes.

79. Microtubular organizing center:

(A) From which microtubules originate.
(B) Rich in gamma tubulin.
(C) Contain two perpendicular centrioles.
(D) All of the above.

80. Liver cells:

(A) May be binucleated.
(B) Contain multiple small fat droplets.
(C) Rich in sER.
(D) All of the above.

81. Fibrillar components of the nuclear membrane includes:

(A) Euchromatin + Lamins.
(B) Lamins.
(C) Heterochromatin + Nuclear lamina.
(D) Lamins + nuclear lamina.

82. All of the following are true as regard the centrioles except:

(A) Present in microtubular organizing center.
(B) Two in non-dividing cells.
(C) Important for mitotic spindle formation.
(D) Contain 27 microfilaments.

83. Enzymes of peroxisomes are formed in:

(A) rER.
(B) Free ribosomes.
(C) Attached ribosomes.
(D) Free & attached ribosomes.

84. Nucleus of the nerve cells:

(A) Condensed.
(B) Euchromatic.
(C) No visible nucleolus.
(D) Inactive.

85. Chromatin:

(A) Basophilic as it contains phosphate group.
(B) Acidophilic as it contains DNA.
(C) Basophilic as it contains protein.
(D) Acidophilic as it contains protein.

75. a 76. a 77. d 78. a 79. d 80. d 81. c 82. d 83. b 84. b 85. a

PRACTICE TIME

1. Which of the following glands eliminates its complete cells as a secretory product?

Ⓐ Merocrine. Ⓑ Apocrine. Ⓒ Holocrine. Ⓓ Endocrine.

2. Which of the following epithelium may have keratin on the surface:

Ⓐ Simple squamous epithelium.
Ⓑ Stratified squamous epithelium.
Ⓒ Simple columnar epithelium.
Ⓓ Transitional epithelium.

3. Which of the following epithelia may have a striate border on its surface:

Ⓐ Simple squamous epithelium.
Ⓑ Stratified squamous epithelium.
Ⓒ Simple columnar epithelium.
Ⓓ Transitional epithelium.

4. Which of the following epithelia is also called the urothelium:

Ⓐ Simple squamous epithelium.
Ⓑ Stratified squamous epithelium.
Ⓒ Simple columnar epithelium.
Ⓓ Transitional epithelium.

5. The type of secretion in which cells lose part of their cytoplasm is called:

Ⓐ Merocrine. Ⓑ Apocrine. Ⓒ Holocrine. Ⓓ Cytocrine.

6. When the cell is expelled as a secretory product, which term applies:

Ⓐ Merocrine. Ⓑ Apocrine. Ⓒ Holocrine. Ⓓ Cytocrine.

7. The respiratory epithelium is also known as:

Ⓐ Endothelium.
Ⓑ Urothelium.
Ⓒ Pseudostratified columnar ciliated with goblet cells.
Ⓓ Mesotelium.

8. Which feature is characteristic for zonula Occludens?

Ⓐ Fusion of adjacent membranes at certain points.
Ⓑ Abundant cytokeratin filaments near the junction.
Ⓒ Surrounds columnar cells in the basal region of their lateral plasma membranes.
Ⓓ Dense intracellular plaque.

9. Stereocilia are

Ⓐ Motile cilia. Ⓑ Short cilia. Ⓒ Short microvilli. Ⓓ Long microvilli.

10. What is a gland called if it has a branched duct?

Ⓐ Simple gland. Ⓑ Compound gland. Ⓒ Tubular. Ⓓ Alveolar.

11. Embryonic ectoderm develops into which of the following major tissue types?

Ⓐ Connective and epithelial tissue.
Ⓑ Connective and muscle tissue.
Ⓒ Muscle tissue and epithelial tissue.
Ⓓ Epithelial tissue and nervous tissue.

1. c 2. b 3. c 4. d 5. b 6. c 7. c 8. a 9. d 10. b 11. d

12. What type of epithelium forms the epidermis?

(A) Stratified squamous epithelium.
(B) Stratified cuboidal epithelium.
(C) Stratified columnar epithelium.
(D) Keratinized stratified squamous epithelium.

13. When the cell is expelled as a secretory product, which term applies:

(A) Merocrine. (B) Apocrine. (C) Holocrine. (D) Cytocrine.

14. In light microscopy, the basement membrane is stained with:

(A) Sudan III. (B) Hematoxylin. (C) Eosin. (D) PAS.

15. The primary purpose of stratification, or layering, in epithelial tissue is for increased _____:

(A) Secretion.
(B) Absorption.
(C) Protection.
(D) Thickening of the basement membrane.

16. What cell surface modification is made of microtubules?

(A) Microvilli. (B) Stereocilia. (C) Cilia. (D) Keratinization.

17. The classical description of a junctional complex includes the following except:

(A) Fascia occludens.
(B) Zonula occludens.
(C) Zonula adherens.
(D) Macula adherens (desmosome).

18. In epithelia, a junctional complex

(A) Is the same as desmosome.
(B) Consists of two adjacent sets of internal laminae.
(C) Generally consists of three regions: zonula occludens, zonula adherens, and macula adherens.
(D) Usually contains a hemidesmosome.

19. Gap junctions

(A) Permit the passage of large proteins from cell to cell.
(B) Form part of the classical junctional complex.
(C) Exist only between epithelial cells.
(D) Are areas of transmission of impulses.

20. The following can be found on the apical surface of intestinal simple columnar epithelium

(A) Villi. (B) Hemidesmosome. (C) Desmosome. (D) Microvilli.

21. What type of epithelium lines most of the gastrointestinal tract?

(A) Simple squamous epithelium.
(B) Simple cuboidal epithelium.
(C) Simple columnar epithelium.
(D) Simple columnar ciliated.

12. d 13. c 14. d 15. c 16. c 17. a 18. c 19. d 20. d 21. c

22. What type of epithelium is composed of cells which all touch the basement membrane and is only one cell layer thick?

- (A) Stratified squamous epithelium.
- (B) Transitional epithelium.
- (C) Stratified cuboidal epithelium.
- (D) Pseudostratified epithelium.

23. A type of epithelium with limited distribution in the body is

- (A) Stratified cubical epithelium.
- (B) Simple columnar epithelium.
- (C) Transitional epithelium.
- (D) Simple squamous epithelium.

24. The stomach is lined by:

- (A) Simple cubical epithelium.
- (B) Pseudostratified columnar ciliated epithelium with goblet cells.
- (C) Simple columnar epithelium.
- (D) Stratified columnar epithelium.

25. Pseudostratified columnar non ciliated epithelium lines:

- (A) Skin.
- (B) Trachea.
- (C) Ureter.
- (D) Vas deferens.

26. Ducts of sweat glands are lined by:

- (A) Simple squamous epithelium.
- (B) Stratified cubical epithelium.
- (C) Simple columnar epithelium.
- (D) Simple cubical epithelium.

27. Regarding transitional epithelium, the following statement is correct:

- (A) Formed of 5-30 layers of cells.
- (B) Cells rest on thick basement membrane.
- (C) Superficial cells are connected by junctional complex.
- (D) Superficial cells are covered by cilia.

28. The epithelial tissue is characterized by:

- (A) Formed of widely separated cells.
- (B) Avascular.
- (C) Great amount of intercellular matrix.
- (D) Arise only from mesoderm.

29. Regarding myo-epithelium, one of the following statements is correct:

- (A) Present in thyroid gland.
- (B) The cells act as a receptor.
- (C) Present between the secretory cells and the duct lumen.
- (D) The cells can contract.

30. Regarding the cilia, choose the correct statement:

- (A) The core is rich in actin filaments.
- (B) The axoneme is formed of 20 microtubules.
- (C) Its main function is increasing the surface area of cells.
- (D) The basal body is formed of 9 doublets of microtubules.

22. d 23. a 24. c 25. d 26. b 27. c 28. b 29. d 30. b

31. Choose the correct statement (junctions):

Ⓐ Desmosomes are maximum at areas of minimal friction.
Ⓑ Tight junction restricts the movement of substance between cells.
Ⓒ Zonula adherens allow the passage of ions between the cells.
Ⓓ The adjacent cell membranes in desmosomes are tightly connected.

32. Regarding nexus (gap junction), choose the correct statement:

Ⓐ It is found in cardiac muscle.
Ⓑ It prevents the passage of substances between cells.
Ⓒ The intercellular space is wide.
Ⓓ It holds the cells together and prevent their separation.

33. Regarding the basement membrane, choose the correct statement:

Ⓐ Basal lamina has no collagen.
Ⓑ Reticular lamina is formed of reticular fibers and ground substance.
Ⓒ It is best seen by Janus green.
Ⓓ The anchoring fibers are represented by collagen type I.

34. Collagen in the basal lamina:

Ⓐ Type I fibers. Ⓑ Type III fibers. Ⓒ Type IV fibers. Ⓓ Type IV granules.

35. Only one statement about the stratified squamous epithelium is correct. Cells of the layer:

Ⓐ Farthest from the basement membrane, squamous.
Ⓑ Farthest from the basement membrane, are polygonal.
Ⓒ Closest to the basement membrane, are squamous.
Ⓓ Closest to the basement membrane are polygonal.

36. The most common type of secretion of exocrine cells is:

Ⓐ All possible types of secretion are roughly equally frequent.
Ⓑ Merocrine secretion.
Ⓒ Holocrine secretion.
Ⓓ Apocrine secretion.

37. Where are usually myoepithelial cells?

Ⓐ In secretory part of glands between secretory cells & the basement membrane.
Ⓑ In secretory part of glands as the layer which is closest to the lumen.
Ⓒ In secretory part of glands present between secretory cells.
Ⓓ In secretory part of glands out of the basement membrane.

38. Which of the following statements is the best description of zonula Occludens? Cell adhesion involved mainly in:

Ⓐ A barrier to avoid free diffusion along epithelial cells.
Ⓑ Exchange of small molecules and ions between cells.
Ⓒ Anchoring the cell to the basement membrane.
Ⓓ Mechanical stability and resistance.

31. b 32. a 33. b 34. d 35. a 36. b 37. a 38. a

39. Cilia:

(A) Are branched forms of microvilli.
(B) Are motile structures on the surface of some epithelial cells.
(C) Are inserted into basement membrane.
(D) Have 9+2 microtubule arrangement.

40. Mesothelium is:

(A) Simple squamous epithelium that lines serous cavities.
(B) Simple squamous epithelium that lines the lumen of the cardiovascular system.
(C) Found in kidney tubules.
(D) B and C can be correct.

41. The cells which specialized to protect underlying tissues from the hypertonic and potentially cytotoxic effects of urine are:

(A) Keratinized cells.
(B) Non-keratinized cells.
(C) Dome cells.
(D) None of the above.

42. Metaplasia is:

(A) Abnormal change in the type of epithelium.
(B) Abnormal growth of the cells.
(C) The same as dysplasia.
(D) B and C are correct.

43. Why is there transitional epithelial tissue in the bladder?

(A) Because this tissue promotes the flow of liquids.
(B) Because this tissue can swell and shrink depending on the fullness of the bladder.
(C) Because this tissue aids in urine absorption.
(D) Because this tissue does not absorb liquids.

44. All, thin epithelial cells that have the appearance of two or more layers, but with all cells touching the basement membrane is

(A) Simple squamous epithelium.
(B) Simple cuboidal epithelium.
(C) Simple columnar epithelium.
(D) Pseudostratified columnar epithelium.

45. Epithelial cells that can stretch from a cuboidal or columnar shape to a squamous shape are called

(A) Simple squamous epithelium.
(B) Simple cuboidal epithelium.
(C) Pseudostratified columnar epithelium.
(D) Transitional epithelium.

46. In which of these locations would pseudostratified columnar ciliated epithelium intermixed with goblet cells be found?

(A) Surface of the skin.
(B) Lining of nasal cavity and trachea.
(C) Lining of serous membranes.
(D) Lining the small intestine.

47. The various types of epithelium are classified by the

(A) Size and shape of cells.
(B) Shape of cells and number of cell layers.
(C) Number of cell layers and size of the cells.
(D) Size and location of cells.

39. b 40. a 41. c 42. a 43. b 44. d 45. d 46. b 47. b

48. Epithelial tissue is distinguished from connective tissue, muscular, or nervous tissue by its

Ⓐ Large extracellular matrix.
Ⓑ Contractility.
Ⓒ Ability to carry action potentials.
Ⓓ Basement membrane.

49. Epithelial tissue that can resist friction would have many

Ⓐ Desmosomes.
Ⓑ Gap junctions.
Ⓒ Tight junctions.
Ⓓ Basement junctions.

50. Glands whose ducts have many branches are called

Ⓐ Simple.
Ⓑ Compound.
Ⓒ Branched.
Ⓓ Alveolar.

51. In a _____ gland, entire cells break down to form the secretion.

Ⓐ Exocrine.
Ⓑ Merocrine.
Ⓒ Holocrine.
Ⓓ Autocrine.

52. According to their function, there are two major types of epithelial tissue, these are

Ⓐ Epithelial tissue and cartilage.
Ⓑ Holocrine and apocrine glands.
Ⓒ Transitional and pseudostratified epithelium.
Ⓓ Covering and secretory epithelium.

53. The secretions of endocrine glands are released directly

Ⓐ Onto the skin surface.
Ⓑ Into the bloodstream.
Ⓒ Into a gland duct.
Ⓓ Into the brain tissue.

54. Which type of epithelium is best suited for rapid filtration?

Ⓐ Simple squamous.
Ⓑ Stratified squamous.
Ⓒ Pseudostratified.
Ⓓ Stratified cuboidal.

55. Which structure forms the brush border?

Ⓐ Secretory granules.
Ⓑ Basal body.
Ⓒ Microvilli.
Ⓓ Cilia.

56. An epithelial tissue having thin flat and closely packed cells that are arranged edge to edge are found in:

Ⓐ The inner lining of blood vessels.
Ⓑ The inner lining of the stomach.
Ⓒ The inner lining of the fallopian tube.
Ⓓ The inner surface of urinary bladder.

57. Tissue having very less or no intercellular matrix is

Ⓐ Epithelial.
Ⓑ Connective.
Ⓒ Cardiac.
Ⓓ Muscular.

58. The inner lining of the vagina, tip of urethra and esophagus is made up of

Ⓐ Columnar epithelium.
Ⓑ Stratified squamous epithelium.
Ⓒ Squamous epithelium.
Ⓓ Ciliated epithelium.

48. d 49. a 50. b 51. c 52. d 53. b 54. a 55. c 56. a 57. a 58. b

59. The basement membrane, one of the following is false:

(A) Stains positively with (PAS).
(B) Is composed of type IV collagen.
(C) Has basal lamina.
(D) Is produced by the connective tissue alone.

60. Transitional epithelium, one of the following is false:

(A) Lines the urinary bladder.
(B) Contains goblet cells.
(C) Is classified as a stratified epithelium.
(D) Has a non-clear basement membrane.

61. Each of the following statements concerning the stratified squamous nonkeratinized epithelium is true, EXCEPT:

(A) Contains multiple cell layers.
(B) Contains the basal layer with columnar cells.
(C) Contains the superficial layer of flat cells.
(D) Lines the oral cavity, the esophagus, the rectum, and the vagina.

62. Each of the following statements concerning the transitional epithelium is true, EXCEPT:

(A) Is keratinized.
(B) Consists of the basal, intermediate, and superficial layers.
(C) Changes the number of its layers, depending on the degree of stretching.
(D) Its superficial cells are dome shaped.

63. Which type of epithelium are cells which are main site of gas exchange?

(A) Pseudostratified columnar epithelium.
(B) Simple cuboidal epithelium.
(C) Simple squamous epithelium.
(D) Transitional epithelium.

64. Which organ contains stratified squamous epithelium with keratinization?

(A) Oral cavity. (B) Esophagus. (C) Trachea. (D) Skin.

65. Which type of collagen is the main part of basal lamina?

(A) Type I collagen.
(B) Type II collagen.
(C) Type III collagen.
(D) Type IV collagen.

66. Which type of epithelium is present epidermis?

(A) Stratified squamous epithelium without keratinization.
(B) Stratified squamous epithelium with keratinization.
(C) Secretory type of epithelium.
(D) Absorptive type of epithelium.

59. d 60. b 61. d 62. a 63. c 64. d 65. d 66. b

67. Epithelium of the bronchi is

(A) Cuboidal.
(B) Simple cuboidal.
(C) Simple squamous.
(D) Pseudo stratified columnar ciliated columnar.

68. How is a pseudostratified epithelium different from a truly stratified tissue?

(A) The nuclei are at the same level in pseudostratified epithelia.
(B) All cells reach apex.
(C) There are multiple cell layers in pseudostratified epithelia.
(D) All the cells in a pseudostratified epithelium interact with the basement membrane.

69. The microvilli containing cells in functional aspect are:

(A) Covering.
(C) Secretory.
(B) Stem.
(D) Absorption.

70. Pseudostratified columnar epithelium is evident in:

(A) Ureter.
(C) Small intestine.
(B) Vas deference.
(D) Esophagus.

71. The epithelial cells connect to the basement membrane through:

(A) Gap junction.
(C) Hemi-desmosomes.
(B) Desmosomes.
(D) Zonula occludens.

72. Which of the following specializations are described in the stratified epithelia:

(A) Microvilli.
(C) Desmosomes.
(B) Cilia.
(D) Gap junction.

67. d 68. b 69. d 70. b 71. c 72. c

PRACTICE TIME

1. Which cell is a connective tissue macrophage?

Ⓐ Kupffer cells. Ⓑ Histiocyte. Ⓒ Dust cell. Ⓓ Langerhans cell.

2. Which of the following can be classified as "specialized connective tissue"?

Ⓐ Mesenchyme.
Ⓑ Mucoid connective tissue.
Ⓒ Dense connective tissue.
Ⓓ Blood.

3. Which of the following is NOT a fiber found in connective tissue?

Ⓐ Collagen fiber. Ⓑ Elastic fiber. Ⓒ Reticular fiber. Ⓓ Purkinje fiber.

4. What type of tissue makes up the dermis of the skin?

Ⓐ Mucous connective tissue.
Ⓑ Reticular connective tissue.
Ⓒ Dense regular connective tissue.
Ⓓ Dense irregular connective tissue.

5. What type of adipose tissue tends to increase as humans age?

Ⓐ Brown adipose tissue.
Ⓑ White adipose tissue.
Ⓒ Unilocular adipose tissue.
Ⓓ Both b and c.

6. Which of the following would be best suited to differentiate collagen fibers from other fibers?

Ⓐ Mallory trichrome stain.
Ⓑ Hematoxylin and eosin stain.
Ⓒ Sudan stain.
Ⓓ Silver stain.

7. The type of connective tissue present in tendon is called:

Ⓐ White adipose. Ⓑ Brown adipose. Ⓒ White fibrous. Ⓓ Mucoid.

8. Which one of these cells is not a cell type routinely found in loose connective tissue?

Ⓐ Fibroblast. Ⓑ Microglia. Ⓒ Histiocyte. Ⓓ Plasma cell.

9. Which of the following can be classified as "specialized connective tissue"?

Ⓐ Cartilage.
Ⓑ Loose connective tissue.
Ⓒ Mesenchyme.
Ⓓ Dense connective tissue.

10. Which of the following can be classified as "connective tissue proper"?

Ⓐ Cartilage.
Ⓑ Dense irregular connective tissue.
Ⓒ Bone.
Ⓓ Blood.

11. What type of tissue is Wharton's jelly?

Ⓐ Mucoid connective tissue.
Ⓑ Mesenchyme.
Ⓒ Loose irregular connective tissue.
Ⓓ Dense irregular connective tissue.

1. b 2. d 3. d 4. d 5. d 6. a 7. c 8. b 9. a 10. b 11. a

PRACTICE TIME

12. What type of tissue is a tendon composed of?

(A) Mucous connective tissue.
(B) Dense regular connective tissue.
(C) Loose irregular connective tissue.
(D) Dense irregular connective tissue.

13. What does connective tissue develop from?

(A) Mesothelium. (B) Mesenchyme. (C) Mesangial cells. (D) Mesentery.

14. What color do elastic fibers stain with Van Gieson stain?

(A) Red/Orange. (B) Pink/red. (C) Purple/Red. (D) Yellow.

15. Which of the following is NOT primarily composed of connective tissue?

(A) Brain. (B) Organ capsule. (C) Ligament. (D) Areolar tissue.

16. Which connective tissue cell type produces the ground substance in connective tissue?

(A) Fibroblast. (B) Mast cell. (C) Histiocyte. (D) Plasma cell.

17. Which connective tissue cell is derived from B lymphocytes?

(A) Fibroblast. (B) Mast cell. (C) Histiocyte. (D) Plasma cell.

18. Embryonic ectoderm develops into which of the following major tissue types?

(A) Connective and epithelial tissue.
(B) Connective and muscle tissue.
(C) Muscle tissue and epithelial tissue.
(D) Epithelial tissue and nervous tissue.

19. Which of the following cells has metachromatic granules:

(A) Mast cells & Lymphocytes.
(B) Mast cells & basophils.
(C) Mast cells & Plasma cells.
(D) Mast cells.

20. The histiocyte stem cells:

(A) Plasmablast. (B) Monocyte. (C) Lymphocyte. (D) Basophile cells.

21. The most abundant type of collagen in the human body is:

(A) Type I. (B) Type II. (C) Type III. (D) Type IV.

22. Which of the following is not associated with connective tissue?

(A) Tightly packed cells.
(B) Extracellular fibers.
(C) Blood vessels.
(D) Ground substance.

23. Which of the following would be best suited to visualize lipid?

(A) Orcein stain.
(B) Hematoxylin and eosin stain.
(C) Sudan III stain.
(D) Silver stain.

12. b 13. b 14. d 15. a 16. a 17. d 18. d 19. b 20. b 21. a 22. a 23. c

24. Which is the most abundant fiber in connective tissue?

(A) Collagen fiber. (B) Elastic fiber. (C) Reticular fiber. (D) Purkinje fiber.

25. Which connective tissue cell type produces collagen?

(A) Fibroblast.
(B) Connective tissue macrophage.
(C) Histiocyte.
(D) Plasma cell.

26. Which connective tissue cell type secretes histamine?

(A) Fibroblast. (B) Fibrocyte. (C) Mast cell. (D) Plasma cell.

27. Which of the following would be best suited to visualize reticular fibers?

(A) Mallory trichrome stain.
(B) Hematoxylin and eosin stain.
(C) Sudan III stain.
(D) Silver stain.

28. Which of the following functions is not characteristic of connective tissue:

(A) Support for epithelia.
(B) Seat of immune reactions.
(C) Diffusion of nutrient.
(D) Contraction.

29. Which of the following cells has the mission of producing heat:

(A) Plasma cells. (B) White adipocytes. (C) Brown adipocytes (D) Mast cells.

30. Which of the following cells of the connective tissue is richest in rER:

(A) Fibrocytes. (B) Monocytes. (C) Plasma cells. (D) Adipocytes.

31. The Type of C.T. contains all types of C.T. cells & fibers is called:

(A) Loose areolar C.T. (B) Adipose C.T. (C) White fibrous C.T. (D) Mucoid C.T.

32. The following is a character of connective tissue:

(A) Formed of crowded cells.
(B) Mesodermal in origin.
(C) Non-vascular.
(D) Tightly joined cells.

33. The following C.T. cells are a protein secreting cells:

(A) Fat and macrophage cells.
(B) Fat and fibroblast cell.
(C) Macrophage and plasma cells.
(D) Fibroblast and plasma cells.

34. The following is a character of multilocular adipocyte:

(A) Has membrane-bound granule filling the cytoplasm.
(B) Has flattened peripheral nucleus.
(C) Function is heat generation.
(D) Most abundant in white adipose C.T.

24. a 25. a 26. c 27. d 28. d 29. c 30. c 31. a 32. b 33. d 34. c

35. After injury of adult C.T., the following cells differentiate for C.T. repair:

(A) Mast cells & Fibroblasts.
(B) Pericytes & Fibroblasts.
(C) Reticular cells & Histiocytes.
(D) Reticular cells & plasma cells.

36. The type III collagen fibers are secreted by the following cells:

(A) Fibroblasts, macrophage cells.
(B) Fibroblasts & reticular cells.
(C) Fibroblasts & mast cells.
(D) Fibroblasts & plasma cells.

37. A young child was suffering from bleeding gums and non-healing wounds. This disease may be due to the defective synthesis of:

(A) Extracellular matrix.
(B) Collagen fibers.
(C) Reticular fibers.
(D) Elastic fibers.

38. Elastic fibers are characterized by being:

(A) Arranged into bundles.
(B) Stained brown with orcein stain.
(C) Not demonstrated with H&E.
(D) Show transverse striations.

39. After receiving Penicillin injection, a patient suffered from allergic reaction which is probably due to secretion of:

(A) Growth factors by fibroblasts.
(B) Lipid by adipocytes.
(C) Antibodies by plasma cells.
(D) Histamine by mast cells.

40. What type of tissue is a ligament composed of?

(A) Mucous connective tissue.
(B) Loose irregular connective tissue.
(C) Dense irregular connective tissue.
(D) Dense regular connective tissue.

41. What type of adipose tissue tends to decrease as humans age?

(A) Both white & brown adipose tissue.
(B) White adipose tissue.
(C) Unilocular adipose tissue.
(D) Multilocular adipose tissue.

42. What type of tissue makes up the reticular layer of dermis of the skin?

(A) Dense regular connective tissue.
(B) Reticular connective tissue.
(C) Loose irregular connective tissue.
(D) Dense irregular connective tissue.

43. Multilocular fat cells are characterized by:

(A) They have signet ring appearance.
(B) They are heat insulator.
(C) Affected by starvation.
(D) Abundant mitochondria.

44. A section in the tendon shows all the following features except:

(A) Minimal intercellular substance.
(B) Many macrophages and mast cells.
(C) Parallel collagen bundles.
(D) Regular white fibrous connective tissue.

35. b 36. b 37. b 38. b 39. d 40. c 41. d 42. d 43. d 44. b

45. All of the following about pericytes are correct except:

- (A) Can differentiate into macrophages.
- (B) Regulate blood flow through the capillaries.
- (C) Are located along capillaries.
- (D) Can differentiate to endothelial cells and smooth muscle cells.

46. One of the following is true about fibroblasts:

- (A) Can differentiate to endothelial cells and smooth muscle cells.
- (B) Secrete collagen and elastic fibers.
- (C) Large cubical cells.
- (D) Responsible for storage of fat.

47. Vitreous humor of the eye is formed of:

- (A) Yellow elastic C.T.
- (B) Reticular C.T.
- (C) Mucoid C.T.
- (D) Irregular white fibrous C.T.

48. Vital stain is used to detect the following cell:

- (A) Plasma cell.
- (B) Reticular cell.
- (C) Macrophage cell.
- (D) Mast cells.

49. Regarding dense irregular connective tissue:

- (A) It is highly vascular.
- (B) It is highly cellular.
- (C) Its main fiber type is collagen.
- (D) It has a large amount of ground substance.

50. Regarding loose areolar connective tissue:

- (A) It is found below epithelium.
- (B) Condensed with fibers.
- (C) It is avascular.
- (D) It has little ground substance.

51. Regarding fibroblasts all of the following are true except:

- (A) Are rare cells in connective tissue.
- (B) Produce collagen fibers.
- (C) Are the main cell type in dense connective tissue.
- (D) Cannot undergo cell division.

52. Epithelial and connective tissue differ from each other in which of the following characteristics?

- (A) Epithelial tissue contains fibers but connective tissue does not.
- (B) Connective tissue is avascular but epithelial tissue is well-vascularized.
- (C) Cells in epithelial tissue are closely packed, whereas in connective tissue they are not.
- (D) Connective tissue includes tissue that makes up glands, but epithelial tissue does not occur in glands.

53. Brown adipose connective tissue is brown due to:

- (A) Cytochrome oxidase only.
- (B) Cytochrome oxidase and blood vessels.
- (C) Carotenoids.
- (D) Thermogenin.

45. a 46. b 47. c 48. c 49. c 50. a 51. a 52. c 53. b

54. Mitochondria of brown fat cell:

(A) Contains thermogenin protein.
(B) Contain cytochrome.
(C) Affected by hormones.
(D) All of the above.

55. Plasma cells is characterized by:

(A) Cartwheel nucleus + negative Golgi image.
(B) Alternating euchromatin & heterochromatin in nucleus.
(C) Deep basophilic cytoplasm.
(D) All of the above.

56. Choose the correct statement:

(A) White fat cells are affected by drugs & hormones.
(B) Brown fat cells are affected by drugs & hormones.
(C) Both white & brown fat cells are affected by drugs & hormones.
(D) None of the above.

57. Antigen presenting cells are:

(A) Macrophages + reticular cells.
(B) Macrophages + plasma cells.
(C) Macrophages + mast cells.
(D) Macrophages + fibroblasts.

58. Yellow elastic connective tissue could be present in:

(A) Large veins.
(B) Large arteries.
(C) Small veins.
(D) All of the above.

59. H&E could be used to stain:

(A) Collagen fibers + elastic fibers.
(B) Elastic fibers + reticular fibers.
(C) Collagen fibers + reticular fibers.
(D) All of the above.

60. Type IV collagen is in the form of:

(A) Granules.
(B) Fibrils.
(C) Fibers.
(D) Bundles.

61. The strongest type of collagen is:

(A) Type I.
(B) Type II.
(C) Type III.
(D) Type IV.

62. Regarding reticular fibers:

(A) Type III collagen.
(B) Stained with Ag.
(C) Stained with PAS.
(D) All of the above.

63. Reticular connective tissue consists of reticular fibers and:

(A) Mast cells.
(B) Plasma cells.
(C) Fibroblasts.
(D) None of the above.

54. d 55. d 56. d 57. a 58. b 59. a 60. a 61. a 62. d 63. d

PRACTICE TIME

1. Which feature is true for specific granules?

(A) Found in lymphocytes.
(B) Found in monocytes.
(C) May contain collagenase, histaminase, or histamine.
(D) Function as lysosomes.

2. Functions of eosinophils include all except:

(A) Destroy histamine by histaminase enzyme.
(B) Engulf antigen – antibody complex.
(C) Kill parasites.
(D) Produce trephone substance.

3. T lymphocytes:

(A) Have kidney-shaped nucleus.
(B) Highly phagocytic.
(C) Responsible for cellular immunity.
(D) Responsible for humoral immunity.

4. What feature is true for the central region of human platelets?

(A) Contains bundles of microtubules.
(B) Contains the platelet granules.
(C) Termed hyalomere.
(D) Contains the platelet nucleus.

5. Which of the following proteins associated with the erythrocyte plasma membrane is responsible for maintaining the cell's biconcave disk shape?

(A) Keratin. (B) Desmin. (C) Lamins. (D) Spectrin.

6. One of the following is a cause of neutrophilia:

(A) Typhoid fever.
(B) Chronic infection.
(C) Parasitic disease.
(D) Acute pyogenic infection.

7. Adaptations of RBCs to function include:

(A) Flexible cell membrane to squeeze in capillaries.
(B) Lipoprotein cell membrane to squeeze in capillaries.
(C) Rounded edges to help gaseous exchange.
(D) Biconcave surface to pass in branched vessels.

8. Which leukocyte is considered as microphage?

(A) Lymphocytes. (B) Basophils. (C) Neutrophils. (D) Monocytes.

9. Which leukocyte is increased in allergic and parasitic infestation?

(A) Eosinophils. (B) Basophils. (C) Neutrophil. (D) a and b.

10. Which leukocyte is the second most abundant in a peripheral smear of blood?

(A) Lymphocytes. (B) Basophils. (C) Neutrophils. (D) Monocytes.

1. c 2. d 3. c 4. b 5. d 6. d 7. a 8. c 9. d 10. a

11. Which of the following is true for platelets?

(A) 2-4 million/mm3.
(B) 2-4 days life span.
(C) 2-4 µm.
(D) 2 types of granules.

12. Which statement typifies sickle cell anemia?

(A) Spherical cell.
(B) Crenated RBC's.
(C) Abnormal RBC's nucleus.
(D) Mutated DNA encoding Hb.

13. Which of the following is described as having a "central pallor"?

(A) Erythrocytes. (B) Monocytes. (C) Eosinophils. (D) Platelets.

14. Which is the smallest leukocyte?

(A) Neutrophil. (B) Lymphocytes. (C) Monocytes. (D) Eosinophil.

15. Which of the following is NOT a term used for a neutrophil?

(A) Polymorphonuclear leukocyte.
(B) First line of defense.
(C) Microphage.
(D) Macrophage.

16. Which of the following is a granulocyte?

(A) Thrombocyte.
(B) Monocyte.
(C) Lymphocyte.
(D) Polymorphonuclear leukocyte.

17. Which cell can divide

(A) Erythrocytes. (B) Thrombocyte. (C) Lymphocyte. (D) Monocyte.

18. Anaphylactic shock occurs due to degranulation of:

(A) Mast cells & Eosinophils.
(B) Eosinophils & Basophils.
(C) Mast cells & Basophils.
(D) Mast cells & plasma cells.

19. A patient suffers from chronic urinary tract infection on top of urinary bilharziasis, which of the following cells could increase in his blood film:

(A) Neutrophils and eosinophils.
(B) Lymphocytes and neutrophils.
(C) Lymphocytes, monocytes, eosinophils & basophils.
(D) Lymphocytes and monocytes only.

20. Which bone marrow component increases markedly as hematopoietic activity decreases?

(A) Fat cells.
(B) Erythrocyte.
(C) Hematopoietic cords.
(D) Stem cells.

21. Basophils contains receptors for:

(A) Ig E. (B) Ig M. (C) Ig D. (D) Ig A.

11. c 12. d 13. a 14. b 15. d 16. d 17. c 18. c 19. c 20. a 21. a

22. Lymphocytosis develops in:

(A) Acute pyogenic infections.
(B) Chronic infections.
(C) Parasitic infections.
(D) Acute abscess formation.

23. Rouleaux appearance:

(A) Occurs in normal circulation.
(B) Reversible phenomena.
(C) Causes damage of RBCs.
(D) Decrease the erythrocyte sedimentation rate.

24. The most common type of leucocytes is:

(A) Neutrophils. (B) Basophils. (C) Lymphocytes. (D) Monocytes.

25. The largest leucocytes is:

(A) Monocyte.
(B) Large lymphocyte.
(C) Neutrophil.
(D) Natural killer.

26. The first line of defense mechanism is:

(A) Neutrophils. (B) Basophils. (C) Lymphocytes. (D) Monocytes.

27. The second line of defense mechanism is:

(A) Neutrophils. (B) Basophils. (C) Lymphocytes. (D) Monocytes.

28. Cells increased in acute pyogenic infection:

(A) Neutrophils. (B) Basophils. (C) Lymphocytes. (D) Monocytes.

29. Cells increased in chronic infection:

(A) Neutrophils. (B) Monocytes. (C) Lymphocytes. (D) b & c.

30. Cells increased in allergic diseases:

(A) Neutrophils. (B) Basophils. (C) Eosinophils. (D) b & c.

31. Reticulocytes could be demonstrated by:

(A) Vital stain.
(B) Supra vital stain.
(C) Toluidine blue.
(D) Metachromatic stain.

32. Frosted glass appearance is a character of:

(A) Neutrophils. (B) Basophils. (C) Lymphocytes. (D) Monocytes.

33. Barr body could be detected in female:

(A) Neutrophils. (B) Basophils. (C) Lymphocytes. (D) Monocytes.

22. b 23. b 24. a 25. a 26. a 27. c 28. a 29. d 30. d 31. b 32. d 33. a

34. Which leukocyte is the second most abundant in a peripheral smear of blood?

(A) Lymphocytes. (B) Basophils. (C) Neutrophils. (D) Monocytes.

35. One of the following is true about eosinophils:

(A) Secrete eosinophil chemotactic factor.
(B) Have horse-shoe shaped nuclei.
(C) Represent about 60-70% of circulating leucocytes.
(D) Secrete histamine and heparin.

36. As regards B Lymphocytes:

(A) Have a long-life span.
(B) Secrete perforins.
(C) Responsible for humoral immunity.
(D) Migrate to thymus gland for differentiation.

37. One is true about RBCs:

(A) Contain azurophilic granules.
(B) The center is thicker than the periphery.
(C) About 10-12 µm in diameter.
(D) Increase in number in chronic lung disease.

38. Basophils:

(A) Are the least common leucocytes.
(B) Increase in influenza.
(C) Have kidney-shaped nuclei.
(D) Contain coarse shiny red granules.

39. Granulomere of blood platelets contains:

(A) Dense tubular system.
(B) Alpha granules that contain serotonin.
(C) Delta granules that contain ATP.
(D) Lambda granules that contain clotting factor.

40. HIV human Immunodeficiency virus that causes AIDs destroys the following type of T lymphocytes:

(A) Cytotoxic T-cells.
(B) Helper T-cells.
(C) Regulatory (Suppressor) T-cells.
(D) Memory T-cells.

41. Regarding the reticulocyte, the following statement is correct:

(A) Acidophilic cytoplasm with basophilic reticulum.
(B) Its size is 20 µm.
(C) Their % increases in case of allergy.
(D) Its percentage is 60-70% of total leucocytic count.

42. The blood film of a patient suffering from bronchial asthma shows:

(A) Neutrophilia and eosinophilia.
(B) Eosinophilia and basophilia.
(C) Basophilia and monocytosis.
(D) Eosinophilia and lymphocytosis.

34. a 35. b 36. c 37. d 38. a 39. c 40. b 41. a 42. b

43. Natural killer cells express the following surface markers (receptors):

(A) CD4. (B) IgM. (C) CD16. (D) CD8.

44. Platelet ribbon is a part of:

(A) Monocyte. (B) Megakaryocyte. (C) Reticulocyte. (D) Lymphocyte.

45. The largest cell in bone marrow is:

(A) Fat cell. (B) Megakaryocyte. (C) Reticular cell. (D) Fibroblast.

46. Lishman's stain is:

(A) Vital stain. (B) Supra vital stain. (C) Neutral stain. (D) Acidic stain.

47. Demarcation channels is important in ..... Formation.

(A) Platelets. (B) Reticulocytes. (C) Megakaryocytes. (D) Leucocytes.

48. Trephone substance is secreted by:

(A) Neutrophils. (B) Basophils. (C) Lymphocytes. (D) Monocytes.

49. Azurophilic granules:

(A) Considered as lysosomes.
(B) Are specific granules.
(C) Stained red in color.
(D) Contain catalase enzymes.

50. Electron dense Internum is a character of:

(A) Eosinophils. (B) Neutrophils. (C) Basophils. (D) Lymphocytes.

51. Hemoglobin is:

(A) Basic protein.
(B) Acidic protein.
(C) Concentrated in the center of the RBCs.
(D) Electron lucent.

52. Both mast cells & basophils share all of the following except:

(A) IgE surface receptors.
(B) Phagocytic activity.
(C) Metachromatic stained granules.
(D) Secretion of eosinophil chemotactic factor.

53. One of the following has a longest life span

(A) T-lymphocytes.
(B) Natural killer cells.
(C) B-lymphocytes.
(D) Both a & b.

54. One of the following has a shortest life span:

(A) Microphage. (B) Macrophage. (C) T-lymphocytes. (D) B-lymphocytes.

43. c 44. b 45. a 46. c 47. a 48. a 49. a 50. a 51. a 52. b 53. d 54. a

55. Tuberculosis is characterized by:

Ⓐ Lymphocytosis. Ⓑ Monocytosis. Ⓒ Neutropenia. Ⓓ All of the above.

56. Specific granules of neutrophils contains all of the following except:

Ⓐ Phagocytin. Ⓑ Lactoferrins. Ⓒ Collagenase. Ⓓ Myeloperoxidase.

57. One of the following is false:

Ⓐ Eosinophil is phagocytic to the antigen antibody complex.
Ⓑ Neutrophil is phagocytic to bacteria.
Ⓒ Monocyte is phagocytic to bacteria.
Ⓓ Mast cell is phagocytic to bacteria.

58. Eosinophils granules include all of the following except:

Ⓐ Neurotoxins to parasites.
Ⓑ Histaminase.
Ⓒ Sulfatase.
Ⓓ Leukotrienes.

59. Central pallor in RBCs is the:

Ⓐ Middle 1/3. Ⓑ Middle 1/2. Ⓒ Middle 2/3. Ⓓ Middle 1/4.

60. Frosted glass appearance of monocytes is due to:

Ⓐ Azurophilic granules.
Ⓑ Specific granules.
Ⓒ Residual bodies.
Ⓓ Phagocytosed particles.

61. As regard demarcation channels in megakaryocytes all of the following are true except:

Ⓐ They are cell membrane invaginations.
Ⓑ Will divide the platelet ribbon into platelets.
Ⓒ Represent cell junctions.
Ⓓ Present in megakaryocytes pseudopodia.

62. Natural killer cells differ from T lymphocytes in:

Ⓐ Have CD16 surface marker.
Ⓑ Being cytotoxic cell.
Ⓒ Born in bone marrow.
Ⓓ Differ in mother cell.

63. T & B lymphocytes can be differentiated with:

Ⓐ LM. Ⓑ EM. Ⓒ Surface markers. Ⓓ Especial stains.

64. Basic protein:

Ⓐ Internum of eosinophils.
Ⓑ Externum of eosinophils.
Ⓒ Induces pores in parasites.
Ⓓ a & c.

65. Lishman stain is:

Ⓐ Neutral stain.
Ⓑ Vital stain.
Ⓒ Supravital stain.
Ⓓ Metachromatic stain.

55. d 56. d 57. d 58. d 59. a 60. a 61. c 62. a 63. c 64. d 65. a